```
mysql> CREATE DATABASE college;
Query OK, 1 row affected (0.04 sec)

mysql> USE college;
Database changed
mysql>
mysql> CREATE TABLE student (
    ->     id INT,
    ->     name VARCHAR(50),
    ->     age INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> ALTER TABLE student ADD email VARCHAR(100);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> ALTER TABLE student MODIFY name VARCHAR(100);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> TRUNCATE TABLE student;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DROP TABLE student;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> USE college;
Database changed
mysql>
mysql> CREATE TABLE department (
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(50) UNIQUE NOT NULL
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     age INT CHECK (age >= 18),
    ->     city VARCHAR(50) DEFAULT 'Chennai',
    ->     dept_id INT,
    ->     email VARCHAR(100) UNIQUE,
    ->     FOREIGN KEY (dept_id) REFERENCES department(dept_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO department VALUES
    -> (1, 'CSE'),
    -> (2, 'IT'),
    -> (3, 'ECE');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO student (id, name, age, dept_id, email)
    -> VALUES
    -> (101, 'Arun', 20, 1, 'arun@gmail.com'),
    -> (102, 'Priya', 21, 2, 'priya@gmail.com'),
    -> (103, 'Kumar', 22, 3, 'kumar@gmail.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM student;
+-----+-------+------+---------+---------+-----------------+
| id  | name  | age  | city    | dept_id | email           |
+-----+-------+------+---------+---------+-----------------+
| 101 | Arun  |   20 | Chennai |       1 | arun@gmail.com  |
| 102 | Priya |   21 | Chennai |       2 | priya@gmail.com |
| 103 | Kumar |   22 | Chennai |       3 | kumar@gmail.com |
+-----+-------+------+---------+---------+-----------------+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT name, age FROM student;
+-------+------+
| name  | age  |
+-------+------+
| Arun  |   20 |
| Priya |   21 |
| Kumar |   22 |
+-------+------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM student WHERE age > 20;
+-----+-------+------+---------+---------+-----------------+
| id  | name  | age  | city    | dept_id | email           |
+-----+-------+------+---------+---------+-----------------+
| 102 | Priya |   21 | Chennai |       2 | priya@gmail.com |
| 103 | Kumar |   22 | Chennai |       3 | kumar@gmail.com |
+-----+-------+------+---------+---------+-----------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM department;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | CSE       |
|       3 | ECE       |
|       2 | IT        |
```

```
mysql> UPDATE student
    -> SET age = 23
    -> WHERE id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE student
    -> SET city = 'Coimbatore'
    -> WHERE id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM student;
+-----+-------+-----+-----------+---------+-----------------+
| id  | name  | age | city      | dept_id | email           |
+-----+-------+-----+-----------+---------+-----------------+
| 101 | Arun  |  23 | Chennai   |       1 | arun@gmail.com  |
| 102 | Priya |  21 | Coimbatore |       2 | priya@gmail.com |
| 103 | Kumar |  22 | Chennai   |       3 | kumar@gmail.com |
+-----+-------+-----+-----------+---------+-----------------+
3 rows in set (0.00 sec)

mysql>
mysql> DELETE FROM student
    -> WHERE id = 103;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM student;
+-----+-------+-----+-----------+---------+-----------------+
| id  | name  | age | city      | dept_id | email           |
+-----+-------+-----+-----------+---------+-----------------+
| 101 | Arun  |  23 | Chennai   |       1 | arun@gmail.com  |
| 102 | Priya |  21 | Coimbatore |       2 | priya@gmail.com |
+-----+-------+-----+-----------+---------+-----------------+
2 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO student (id, name, age, dept_id, email)
    -> VALUES (104, 'Ravi', 20, 1, 'ravi@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE student
    -> SET age = 25
    -> WHERE id = 104;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM student;
+-----+-------+-----+-----------+---------+-----------------+
| id  | name  | age | city      | dept_id | email           |
+-----+-------+-----+-----------+---------+-----------------+
| 101 | Arun  |  23 | Chennai   |       1 | arun@gmail.com  |
| 102 | Priya |  21 | Coimbatore|       2 | priya@gmail.com |
| 104 | Ravi  |  20 | Chennai   |       1 | ravi@gmail.com  |
+-----+-------+-----+-----------+---------+-----------------+
3 rows in set (0.00 sec)

mysql>
```